

A Control-Anchored Mean-Field Neural Differential Equation Formulation for P4-to-P60 Perturb-seq Dynamics

Abstract

We formulate the $P4 \rightarrow P60$ perturb-seq problem as inference of a perturbation-indexed family of finite-measure-valued stochastic processes on a latent cell-state space. The biological motivation is that a perturbation can alter at least three coupled aspects of tumour evolution: directional progression through cell-state space, stochastic heterogeneity around that progression, and net expansion or depletion of the perturbed population. Because single-cell RNA sequencing is destructive, the observed data are snapshot distributions rather than tracked single-cell trajectories. This motivates a weighted stochastic differential equation formulation in a shared latent space. For each perturbation $g \in \mathcal{G}$, the observed $P4$ cells and guide-abundance scale define an initial finite measure $\mu_0^g = M_g^0 p_0^g$, while the observed $P60$ cells and guide-abundance scale define a terminal finite measure $\nu_1^g = M_g^1 p_1^g$. The model is control anchored: control perturbations are represented by the zero embedding and therefore evaluate the baseline coefficient fields of the same networks. The model is also mean-field coupled: perturbation-specific populations jointly determine a low-dimensional tissue context, and this context feeds back into the perturbation-conditioned dynamics. Mathematically, the resulting system is a reaction–drift–diffusion McKean–Vlasov model in latent space, with three separate control-anchored neural networks for directional movement, stochastic dispersion, and multiplicative growth or depletion. The theoretical object is a law-level family of finite measures, while the weighted particle system is a numerical approximation to that family. The predicted terminal finite measures are fitted to data with a single entropic unbalanced optimal-transport discrepancy, because terminal geometry and terminal abundance are two aspects of the same endpoint measure rather than two independent targets. Coarse cell-population labels may be used to build interpretable tissue-context summaries, but they are not required for the core endpoint supervision. The learned object is an effective interval generator for the single observed interval $P4 \rightarrow P60$; it is not claimed to be the unique microscopic biological law between the two snapshots.

Contents

1	Biological background and modeling target	3
1.1	Biological setting	3
1.2	Why the data imply a population model rather than a trajectory-matching model . .	4
1.3	Why control anchoring is natural	4
1.4	Why mean-field tissue context is biologically plausible	4
1.5	What the model should estimate	4
2	Explicit introduction to the weighted SDE	5
2.1	What an SDE is	5
2.2	Why an ODE alone is not enough	5

2.3	Why we also need a growth equation	5
2.4	The weighted SDE we actually want	6
2.5	From a representative trajectory to a finite measure	6
3	Notation table	6
4	Observed data and endpoint object	8
4.1	Normalized time	8
4.2	Latent state space	8
4.3	Perturbation-indexed single-cell observations	8
4.4	Guide-abundance masses and finite endpoint measures	9
4.5	Practical support requirement	9
4.6	Optional interpretable tissue summaries	9
5	Assumptions	10
6	Control anchoring and mean-field tissue context	10
6.1	Control perturbations and perturbation embeddings	10
6.2	Joint population and total mass	11
6.3	Generic mean-field context	11
6.4	One interpretable choice of context map	11
6.5	Time embedding and common network input	11
7	Explicit control-anchored three-network weighted SDE	12
7.1	Why three separate networks	12
7.2	Drift network: directional movement	12
7.3	Dispersion network: stochastic spread	12
7.4	Growth network: expansion or depletion	13
7.5	Exact control anchor	13
7.6	Law-level representative process	13
7.7	Weak measure form at the law level	14
7.8	Particle approximation	14
7.9	Empirical weak form and martingale fluctuation	14
7.10	Regularity conditions	15
8	Explicit details behind the loss function	15
8.1	Why the loss must act on finite measures	15
8.2	Derived endpoint summaries	15
8.3	Generalized KL divergence for finite measures	16
8.4	Entropic unbalanced optimal transport	16
8.5	Debiased unbalanced Sinkhorn divergence	16
8.6	Primary endpoint loss	16
8.7	Model-side growth versus loss-side mass relaxation	17
8.8	Optional auxiliary composition diagnostic	17

9	Regularization and identifiability	17
9.1	Embedding shrinkage	17
9.2	Perturbation-modulation regularization	18
9.3	Dispersion regularization	18
9.4	Network parameter regularization	18
9.5	Why strong regularization is necessary	18
10	Full training objective	18
11	Numerical approximation and training	19
11.1	Euler–Maruyama approximation	19
11.2	Context update	19
11.3	Discrete terminal measures	19
11.4	Practical training loop	20
12	Biological outputs and counterfactuals	20
12.1	Predicted terminal finite measure	20
12.2	Predicted abundance effect	20
12.3	Predicted terminal state distribution	20
12.4	Predicted terminal coarse composition	21
12.5	Intrinsic versus context-mediated effect	21
12.6	Control-referenced perturbation effect	21
13	What this model is, and is not	21
13.1	It is a mean-field interaction model	21
13.2	It is not a full mean-field game	21
13.3	It estimates an effective interval generator	21
13.4	It supports intervention-conditional interpretation, not unrestricted causal claims	21
14	Concise summary	22

1 Biological background and modeling target

1.1 Biological setting

The biological problem is to understand how a perturbation reshapes tumour evolution over the observed interval

$$P4 \longrightarrow P60.$$

At a biological level, a perturbation can affect at least three coupled mechanisms.

First, it can change the *direction* in which cells move through transcriptomic state space. A perturbation may promote progression toward one epithelial program, block another program, or redirect cells into a different branch of the state manifold.

Second, it can change the amount of *heterogeneity* or unresolved stochastic variation around that dominant direction. Even if two perturbations have similar average trends, one may produce a broader spread of cell states and another may confine the population more tightly.

Third, it can change the *net abundance* of the perturbed population. A perturbation may confer a selective advantage, cause depletion, alter survival, or change proliferative capacity.

These three effects—movement, stochastic spread, and net growth or loss—are precisely the three components that motivate the mathematical structure of the model below.

1.2 Why the data imply a population model rather than a trajectory-matching model

Single-cell RNA sequencing is destructive. One does not observe the same physical cell at $P4$ and again at $P60$. Therefore, the data do not provide directly tracked cell lineages in state space. Instead, for each perturbation, one observes:

- a collection of cell states at $P4$,
- a collection of cell states at $P60$,
- and guide-abundance information that provides a relative mass scale.

This means that the natural object of inference is not a set of matched cell-to-cell trajectories. The natural object is a *population dynamics model* that evolves a perturbation-specific distribution or finite measure from the $P4$ snapshot to the $P60$ snapshot.

1.3 Why control anchoring is natural

The experimental design contains control perturbations, such as non-targeting guides. These controls provide a natural reference continuation law. Rather than learning a completely unrelated high-capacity dynamics for each perturbation, it is more statistically efficient and biologically interpretable to model each perturbation as a deviation from a shared control law. This is the reason for introducing a perturbation embedding a_g with the exact anchor

$$a_g = 0 \quad \text{for controls.}$$

1.4 Why mean-field tissue context is biologically plausible

A perturbation can act cell-intrinsically, but the evolving tissue composition can also feed back on cell behavior. In a tumour setting, the current mixture of epithelial states, immune states, and signaling-related features can alter the effective continuation law. This motivates a mean-field interaction layer: the coefficient fields are allowed to depend not only on the current cell state and perturbation identity, but also on low-dimensional summaries of the current population.

1.5 What the model should estimate

The target is a perturbation-conditioned family of evolving finite measures

$$\left\{ \mu_t^g : t \in [0, 1], \ g \in \mathcal{G} \right\},$$

where $t = 0$ corresponds to $P4$ and $t = 1$ corresponds to $P60$. The same model should explain both:

- where the perturbed population ends up in latent state space, and
- how much total perturbation-specific mass survives, expands, or depletes.

2 Explicit introduction to the weighted SDE

2.1 What an SDE is

A stochastic differential equation describes how a random state evolves over infinitesimal time. If $Z_t \in \mathbb{R}^d$ denotes a latent cell state, a generic SDE has the form

$$dZ_t = b(Z_t, t) dt + \sigma(Z_t, t) dW_t, \quad (1)$$

where:

- $b(Z_t, t)$ is the *drift*, the deterministic infinitesimal direction of motion;
- $\sigma(Z_t, t)$ is the *diffusion map*, which scales stochastic fluctuations;
- W_t is a Brownian motion, the canonical continuous-time noise process.

Over a small time interval Δt , [equation \(1\)](#) can be read heuristically as

$$Z_{t+\Delta t} - Z_t \approx b(Z_t, t) \Delta t + \sigma(Z_t, t) \sqrt{\Delta t} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I). \quad (2)$$

Thus, the drift term determines the average direction of motion, while the diffusion term determines the size and orientation of stochastic spread around that direction.

2.2 Why an ODE alone is not enough

A deterministic ODE would only include the drift term,

$$dZ_t = b(Z_t, t) dt.$$

That is too restrictive here. In the present problem, the latent state does not capture all biological variables, and the observed data consist only of sparse destructive snapshots. Therefore, a stochastic dispersion term is not merely a nuisance; it is part of the effective interval-level description. It absorbs unresolved heterogeneity, latent-state incompleteness, and intrinsically variable continuation around the main direction of motion.

2.3 Why we also need a growth equation

The SDE in [equation \(1\)](#) moves particles through latent state space, but it does not by itself account for perturbation-specific expansion or depletion. To model abundance changes, we attach a positive weight w_t to each particle and write

$$\frac{d}{dt} \log w_t = r(Z_t, t), \quad (3)$$

where r is a scalar growth or reaction field.

This has the explicit solution

$$w_t = \exp\left(\int_0^t r(Z_s, s) ds\right). \quad (4)$$

If $r > 0$ along the trajectory, the particle weight grows; if $r < 0$, the weight shrinks.

2.4 The weighted SDE we actually want

For perturbation-conditioned tumour evolution, the natural conceptual model is therefore

$$dZ_t = b_g(Z_t, t; \mu_t) dt + \sigma_g(Z_t, t; \mu_t) dW_t, \quad (5)$$

together with

$$\frac{d}{dt} \log w_t = r_g(Z_t, t; \mu_t). \quad (6)$$

The perturbation index g appears because different perturbations modify the continuation law. The dependence on μ_t appears because the current tissue context can feed back on the dynamics through a mean-field summary of the evolving population.

2.5 From a representative trajectory to a finite measure

A single weighted trajectory does not represent the full perturbation-specific population. The theoretical model therefore evolves a *law-level finite measure* μ_t^g obtained by averaging over representative weighted trajectories. The particle system introduced later is a numerical approximation to this law-level object rather than the definition of the model itself.

3 Notation table

Table 1: Core notation used throughout the formulation.

Symbol	Type	Meaning
$\mathcal{G}$	finite set	Set of perturbations.
$\mathcal{G}_{\text{ctrl}}$	subset of $\mathcal{G}$	Set of control perturbations.
g, h	elements of $\mathcal{G}$	Perturbation indices.
$t \in [0, 1]$	scalar	Normalized time, with $t = 0$ for $P4$ and $t = 1$ for $P60$.
$\tau(t)$	scalar	Optional map from normalized time to physical days.
$\mathcal{Z} \subseteq \mathbb{R}^d$	space	Shared latent cell-state space.
x_{gi}, y_{gj}	vectors in $\mathcal{Z}$	Observed latent states of cell i at $P4$ and cell j at $P60$ under perturbation g .
X_g, Y_g	finite sets	Observed latent cell-state collections at $P4$ and $P60$ under perturbation g .
n_g^0, n_g^1	integers	Numbers of observed $P4$ and $P60$ cells under perturbation g .
p_0^g, p_1^g	probability measures	Empirical probability measures of the observed latent states at $P4$ and $P60$.
M_g^0, M_g^1	positive scalars	Relative perturbation masses on the guide-abundance scale at $P4$ and $P60$.
μ_0^g, ν_1^g	finite measures	Initial and observed terminal finite measures, $\mu_0^g = M_g^0 p_0^g$ and $\nu_1^g = M_g^1 p_1^g$.
μ_t^g	finite measure	Law-level predicted perturbation-specific population measure at time t .

Symbol	Type	Meaning
$\hat{\mu}_t^{g,N_g}$	random finite measure	Weighted empirical particle approximation to μ_t^g .
$\bar{\mu}_t$	finite measure	Total law-level population measure $\sum_{g \in \mathcal{G}} \mu_t^g$.
$\bar{M}(t)$	positive scalar	Total mass $\bar{\mu}_t(\mathcal{Z})$.
N_g	integer	Number of particles used numerically for perturbation g .
a_g	vector in $\mathbb{R}^r$	Perturbation embedding; controls satisfy $a_g = 0$.
ω	map $\mathcal{Z} \rightarrow \Delta^{K-1}$	Optional coarse membership map used for interpretable context summaries.
ϕ	map $\mathcal{Z} \rightarrow \mathbb{R}^J$	Optional mediator or signaling feature map.
q_t	vector in Δ^{K-1}	Optional global coarse composition summary.
s_t	vector in $\mathbb{R}^J$	Optional global mediator summary.
c_t	vector in $\mathbb{R}^L$	Low-dimensional mean-field tissue-context vector.
$\hat{c}_t$	vector in $\mathbb{R}^L$	Numerical approximation to c_t computed from particles.
$\gamma(t)$	vector in $\mathbb{R}^{d_t}$	Time embedding.
$u_t(z; \mu_t)$	vector	Shared network input built from latent state, time, and tissue context.
b_g	vector field	Drift field governing directional movement.
σ_g	matrix field	Diffusion map governing stochastic dispersion.
r_g	scalar field	Growth or reaction field governing multiplicative expansion or depletion.
W_t^g	Brownian motion	Brownian motion for a representative perturbation- g trajectory.
$W_t^{g,i}$	Brownian motion	Brownian motion for particle i under perturbation g .
Z_t^g	stochastic process	Representative latent state trajectory under perturbation g .
$Z_t^{g,i}$	stochastic process	Latent state trajectory of particle i under perturbation g .
w_t^g	positive process	Representative perturbation- g particle weight.
$w_t^{g,i}$	positive process	Weight of particle i under perturbation g .
$\hat{M}_g^1$	scalar	Predicted terminal mass $\mu_1^g(\mathcal{Z})$.
$\hat{p}_1^g$	probability measure	Predicted terminal normalized state distribution $\mu_1^g / \hat{M}_g^1$.
$\hat{q}_{g,1}$	vector in Δ^{K-1}	Optional predicted terminal coarse composition.
$\sigma_{\min}$	positive scalar	Lower bound used in the diffusion parameterization.
$r_{\max}$	positive scalar	Bound used in the growth parameterization.
ε	positive scalar	Entropic regularization parameter in unbalanced OT.
τ	positive scalar	Marginal-relaxation parameter in unbalanced OT.
$\text{UOT}_{\varepsilon,\tau}$	functional	Entropically regularized unbalanced OT cost on finite measures.

Symbol	Type	Meaning
$\text{SD}_{\varepsilon, \tau}^{\text{unb}}$	functional	Debiased unbalanced Sinkhorn divergence on finite measures.
$\theta_b, \theta_\sigma, \theta_r, \theta_c$	parameters	Parameters of the drift, dispersion, growth, and context networks.

4 Observed data and endpoint object

4.1 Normalized time

We identify

$$t = 0 \equiv P4, \quad t = 1 \equiv P60.$$

If one wishes to connect normalized time to physical days, one may set

$$\tau(t) = 4 + 56t,$$

so that $\tau(0) = 4$ and $\tau(1) = 60$.

4.2 Latent state space

Let

$$\mathcal{Z} \subseteq \mathbb{R}^d$$

denote a common latent cell-state space obtained from pooled $P4$ and $P60$ transcriptomes. This latent space may be PCA, scVI, or another shared representation. The working assumption is that $\mathcal{Z}$ is sufficiently informative for an effective interval-level dynamics on $[0, 1]$.

4.3 Perturbation-indexed single-cell observations

For each perturbation $g \in \mathcal{G}$, let

$$X_g = \{x_{g1}, \dots, x_{gn_g^0}\} \subset \mathcal{Z}$$

be the latent states of observed $P4$ cells carrying perturbation g , and let

$$Y_g = \{y_{g1}, \dots, y_{gn_g^1}\} \subset \mathcal{Z}$$

be the latent states of observed $P60$ cells carrying perturbation g .

Define the empirical probability measures

$$p_0^g = \frac{1}{n_g^0} \sum_{i=1}^{n_g^0} \delta_{x_{gi}}, \quad p_1^g = \frac{1}{n_g^1} \sum_{j=1}^{n_g^1} \delta_{y_{gj}}. \quad (7)$$

4.4 Guide-abundance masses and finite endpoint measures

The numbers of sequenced cells and the guide-abundance measurements need not be on the same scale. Accordingly, we distinguish:

- *state samples*: n_g^0 and n_g^1 , used to define p_0^g and p_1^g ;
- *guide-abundance masses*: $M_g^0 > 0$ and $M_g^1 > 0$, representing normalized relative perturbation abundance at $P4$ and $P60$ on a guide-abundance scale.

The initial and observed terminal finite measures are

$$\mu_0^g = M_g^0 p_0^g, \quad \nu_1^g = M_g^1 p_1^g. \quad (8)$$

This is the key endpoint object. For each perturbation g , the observed terminal target is the *finite measure* ν_1^g , which already combines:

- (i) terminal geometry in latent state space through p_1^g , and
- (ii) terminal abundance on the guide-abundance scale through M_g^1 .

Interpretive caveat. The quantities M_g^0 and M_g^1 should be read as abundance proxies on a normalized guide-abundance scale rather than as exact absolute clone sizes. The finite measures in [equation \(8\)](#) are therefore effective endpoint targets. If one wants to propagate mass uncertainty explicitly, one may introduce latent true masses together with an observation model for the abundance readout, but that extension is not required for the core formulation here.

4.5 Practical support requirement

The formulas in [equation \(7\)](#) assume $n_g^0, n_g^1 > 0$. In practice, perturbations with extremely small transcriptomic support should be filtered, pooled across sgRNAs targeting the same gene, or handled with a partial observation model that uses guide abundance without a full state-resolved endpoint distribution. The exposition below assumes sufficient single-cell support at both endpoints for the perturbations that are modeled.

4.6 Optional interpretable tissue summaries

If one wants a biologically interpretable mean-field context, one may define:

$$\omega : \mathcal{Z} \rightarrow \Delta^{K-1}, \quad \phi : \mathcal{Z} \rightarrow \mathbb{R}^J,$$

where ω is an optional coarse population membership map and ϕ is an optional mediator or signaling feature map.

For each perturbation g , the observed terminal coarse composition is then

$$q_{g,1}^{\text{obs}} = \frac{1}{n_g^1} \sum_{j=1}^{n_g^1} \omega(y_{gj}) = \frac{1}{M_g^1} \int_{\mathcal{Z}} \omega(z) \nu_1^g(dz). \quad (9)$$

These summaries are useful for interpretation, but they are not required for the core endpoint loss. The core supervision acts directly on finite measures.

5 Assumptions

The formulation rests on the following explicit assumptions.

- (A1) **Latent Markov sufficiency.** The pooled latent representation $\mathcal{Z}$ is sufficiently informative that the $P4 \rightarrow P60$ interval can be described by an effective Markovian dynamics on $\mathcal{Z}$.
- (A2) **Within-perturbation exchangeability.** For each perturbation g , conditional on latent state and the shared tissue context, the joint law of cells carrying g is invariant under permutation of cell indices. Thus, cells with the same perturbation are modeled as interchangeable particles, and the perturbation-specific population law is the sufficient population-level state.
- (A3) **Relative-mass interpretation.** The masses M_g^0 and M_g^1 are treated as guide-abundance-scale abundance proxies and are therefore comparable across perturbations after normalization, even though they are distinct from the number of sequenced single-cell transcriptomes.
- (A4) **Mean-field closure.** The effect of the tissue context on any cell is approximated through low-dimensional aggregate summaries of the joint population rather than through explicit pairwise interactions between all individual cells.
- (A5) **Control anchoring.** Control perturbations define the baseline continuation law at the coefficient level. Perturbation effects are represented as low-dimensional deviations away from that baseline rather than as fully unrelated dynamics for each perturbation.
- (A6) **Effective-interval identifiability.** Because only two state-resolved time points are used, the learned dynamics are interpreted as an effective generator on $[0, 1]$, not as a uniquely identified microscopic law at all intermediate times.
- (A7) **Pooled-context approximation.** When replicate-specific tissue context is not explicitly modeled, the single context vector c_t is interpreted as an effective screen-average environment rather than as the literal within-animal environment experienced by every cell.

6 Control anchoring and mean-field tissue context

6.1 Control perturbations and perturbation embeddings

Let

$$\mathcal{G}_{\text{ctrl}} \subset \mathcal{G}$$

denote the set of control perturbations. Each perturbation $g \in \mathcal{G}$ is assigned an embedding

$$a_g \in \mathbb{R}^r.$$

To make the control anchor exact, we impose

$$a_g = 0 \quad \text{for every } g \in \mathcal{G}_{\text{ctrl}}. \quad (10)$$

Many perturb-seq experiments use multiple sgRNAs for the same target. In that setting, a hierarchical decomposition is often preferable:

$$a_{gr} = a_g + \delta_{gr}, \quad \delta_{gr} \in \mathbb{R}^r, \quad (11)$$

with shrinkage of δ_{gr} toward zero. For expositional simplicity, the main formulas below use a single perturbation-level embedding a_g .

6.2 Joint population and total mass

Let

$$\mu_t = (\mu_t^g)_{g \in \mathcal{G}}$$

denote the family of perturbation-indexed populations at time t . The total population measure is

$$\bar{\mu}_t = \sum_{g \in \mathcal{G}} \mu_t^g, \quad \bar{M}(t) = \bar{\mu}_t(\mathcal{Z}). \quad (12)$$

6.3 Generic mean-field context

The most general formulation uses a low-dimensional tissue-context vector

$$c_t = \Psi_{\theta_c}(\mu_t) \in \mathbb{R}^L, \quad (13)$$

where Ψ_{θ_c} is a context map from the current joint population law to a low-dimensional summary.

This expression is the precise mathematical content of the mean-field coupling: the coefficient fields are allowed to depend on the current population only through the context vector c_t . The no-context baseline is recovered by setting

$$c_t \equiv 0.$$

6.4 One interpretable choice of context map

One interpretable implementation is to define

$$q_t = \frac{1}{\bar{M}(t)} \sum_{g \in \mathcal{G}} \int_{\mathcal{Z}} \omega(z) \mu_t^g(\mathrm{d}z) \in \Delta^{K-1}, \quad (14)$$

and

$$s_t = \frac{1}{\bar{M}(t)} \sum_{g \in \mathcal{G}} \int_{\mathcal{Z}} \phi(z) \mu_t^g(\mathrm{d}z) \in \mathbb{R}^J. \quad (15)$$

Then set

$$c_t = \Psi_{\theta_c}(q_t, s_t) \in \mathbb{R}^L. \quad (16)$$

This implementation keeps the mean-field layer interpretable in terms of coarse population composition and mediator features, but the core formulation does not require this particular choice.

Sample-specific extension. If the data include distinct mice, tumours, or experimental batches, a more faithful extension is to index populations and contexts by sample s , for example through $(\mu_{s,t}^g)_g$ and $c_{s,t} = \Psi_{\theta_c}(\mu_{s,t})$. The pooled version in [equations \(13\) and \(16\)](#) should then be interpreted as an effective screen-average context.

6.5 Time embedding and common network input

Let

$$\gamma : [0, 1] \rightarrow \mathbb{R}^{d_t}$$

be a time embedding. For a latent state $z \in \mathcal{Z}$, define

$$u_t(z; \mu_t) = (z, \gamma(t), c_t). \quad (17)$$

This is the common input to the coefficient networks.

7 Explicit control-anchored three-network weighted SDE

7.1 Why three separate networks

The conceptual model in [equations \(5\)](#) and [\(6\)](#) has three biologically distinct mechanisms:

- (i) directional movement,
- (ii) stochastic dispersion,
- (iii) multiplicative growth or depletion.

These mechanisms have different mathematical constraints and different identifiability properties. For that reason, they should not be collapsed into a single undifferentiated neural output. We therefore use:

- (1) a *drift network*,
- (2) a *dispersion network*,
- (3) a *growth network*.

7.2 Drift network: directional movement

Let

$$\text{DriftNet}_{\theta_b} : \mathbb{R}^{d+d_t+L} \rightarrow \mathbb{R}^d \times \mathbb{R}^{d \times r},$$

and write

$$\text{DriftNet}_{\theta_b}(u) = (\beta_b(u), B_b(u)),$$

where

$$\beta_b(u) \in \mathbb{R}^d, \quad B_b(u) \in \mathbb{R}^{d \times r}.$$

The perturbation-conditioned drift field is

$$b_g(z, t; \mu_t) = \beta_b(u_t(z; \mu_t)) + B_b(u_t(z; \mu_t)) a_g. \quad (18)$$

The first term is the baseline control law; the second term is the perturbation-specific deviation from control.

7.3 Dispersion network: stochastic spread

Let

$$\text{DispersionNet}_{\theta_\sigma} : \mathbb{R}^{d+d_t+L} \rightarrow \mathbb{R}^d \times \mathbb{R}^{d \times r},$$

and write

$$\text{DispersionNet}_{\theta_\sigma}(u) = (\beta_\sigma(u), B_\sigma(u)),$$

where

$$\beta_\sigma(u) \in \mathbb{R}^d, \quad B_\sigma(u) \in \mathbb{R}^{d \times r}.$$

Define the pre-activation

$$\ell_g(z, t; \mu_t) = \beta_\sigma(u_t(z; \mu_t)) + B_\sigma(u_t(z; \mu_t)) a_g \in \mathbb{R}^d, \quad (19)$$

and then define the diffusion matrix by

$$\sigma_g(z, t; \mu_t) = \text{diag}(\text{softplus}(\ell_g(z, t; \mu_t)) + \sigma_{\min}), \quad \sigma_{\min} > 0. \quad (20)$$

The diagonal form is the default because diffusion is only weakly identifiable from two endpoint snapshots. If additional information becomes available, one may replace this with a low-rank or full-covariance parameterization.

7.4 Growth network: expansion or depletion

Let

$$\text{GrowthNet}_{\theta_r} : \mathbb{R}^{d+d_t+L} \rightarrow \mathbb{R} \times \mathbb{R}^{1 \times r},$$

and write

$$\text{GrowthNet}_{\theta_r}(u) = (\beta_r(u), B_r(u)),$$

where

$$\beta_r(u) \in \mathbb{R}, \quad B_r(u) \in \mathbb{R}^{1 \times r}.$$

Define the unconstrained log-rate

$$\tilde{r}_g(z, t; \mu_t) = \beta_r(u_t(z; \mu_t)) + B_r(u_t(z; \mu_t)) a_g, \quad (21)$$

and then set

$$r_g(z, t; \mu_t) = r_{\max} \tanh(\tilde{r}_g(z, t; \mu_t)), \quad r_{\max} > 0. \quad (22)$$

The bounded tanh form stabilizes training because the particle weights depend exponentially on the time integral of r_g .

7.5 Exact control anchor

Because controls satisfy $a_g = 0$, [equations \(18\), \(19\)](#) and [\(21\)](#) imply that, *conditional on the shared context trajectory c_t* , control perturbations evaluate the baseline coefficient fields

$$\beta_b(\cdot), \quad \beta_\sigma(\cdot), \quad \beta_r(\cdot).$$

This is an exact coefficient-level anchor. It should not be confused with the stronger claim that controls evolve in a perturbation-free global environment, because the shared context c_t can itself depend on the full perturbation mixture.

7.6 Law-level representative process

For each perturbation g , let (Z_t^g, w_t^g) denote a representative weighted trajectory solving

$$dZ_t^g = b_g(Z_t^g, t; \mu_t) dt + \sigma_g(Z_t^g, t; \mu_t) dW_t^g, \quad Z_0^g \sim p_0^g, \quad (23)$$

$$\frac{d}{dt} \log w_t^g = r_g(Z_t^g, t; \mu_t), \quad w_0^g = 1, \quad (24)$$

where W_t^g is a d -dimensional Brownian motion.

The perturbation-specific law-level finite measure is then defined by

$$\int_{\mathcal{Z}} f(z) \mu_t^g(dz) = M_g^0 \mathbb{E}[w_t^g f(Z_t^g)] \quad \text{for every bounded measurable } f : \mathcal{Z} \rightarrow \mathbb{R}. \quad (25)$$

In particular, taking $f \equiv 1$ gives the predicted terminal mass

$$\hat{M}_g^1 = \mu_1^g(\mathcal{Z}) = M_g^0 \mathbb{E}[w_1^g]. \quad (26)$$

At $t = 0$, [equation \(25\)](#) recovers $\mu_0^g = M_g^0 p_0^g$.

7.7 Weak measure form at the law level

For every smooth test function $f : \mathcal{Z} \rightarrow \mathbb{R}$,

$$\frac{d}{dt} \int_{\mathcal{Z}} f(z) \mu_t^g(dz) = \int_{\mathcal{Z}} \left[b_g(z, t; \mu_t) \cdot \nabla f(z) + \frac{1}{2} \text{tr} \left(D_g(z, t; \mu_t) \nabla^2 f(z) \right) + r_g(z, t; \mu_t) f(z) \right] \mu_t^g(dz), \quad (27)$$

where

$$D_g(z, t; \mu_t) = \sigma_g(z, t; \mu_t) \sigma_g(z, t; \mu_t)^\top.$$

If a density exists, [equation \(27\)](#) corresponds formally to

$$\partial_t \mu_t^g = -\nabla \cdot (b_g(\cdot, t; \mu_t) \mu_t^g) + \frac{1}{2} \nabla \cdot \nabla : (D_g(\cdot, t; \mu_t) \mu_t^g) + r_g(\cdot, t; \mu_t) \mu_t^g. \quad (28)$$

7.8 Particle approximation

The weighted particle system is a numerical approximation to the law-level measure μ_t^g . For each perturbation g , let $\{(Z_t^{g,i}, w_t^{g,i})\}_{i=1}^{N_g}$ solve

$$dZ_t^{g,i} = b_g(Z_t^{g,i}, t; \hat{\mu}_t) dt + \sigma_g(Z_t^{g,i}, t; \hat{\mu}_t) dW_t^{g,i}, \quad (29)$$

$$\frac{d}{dt} \log w_t^{g,i} = r_g(Z_t^{g,i}, t; \hat{\mu}_t), \quad w_0^{g,i} = 1, \quad (30)$$

with empirical measure

$$\hat{\mu}_t^{g, N_g} = \frac{M_g^0}{N_g} \sum_{i=1}^{N_g} w_t^{g,i} \delta_{Z_t^{g,i}} \in \mathcal{M}_+(\mathcal{Z}). \quad (31)$$

Here $\hat{\mu}_t = (\hat{\mu}_t^{g, N_g})_{g \in \mathcal{G}}$ denotes the current family of empirical measures.

The simplest implementation takes $N_g = n_g^0$ and initializes the particles at the observed $P4$ cells,

$$Z_0^{g,i} = x_{gi}.$$

Then [equation \(31\)](#) matches the observed initial finite measure exactly at the numerical level.

7.9 Empirical weak form and martingale fluctuation

Because [equation \(31\)](#) is a random empirical measure, its dynamics are not purely deterministic.

For every smooth test function $f : \mathcal{Z} \rightarrow \mathbb{R}$,

$$d \int_{\mathcal{Z}} f(z) \hat{\mu}_t^{g, N_g}(dz) = \left[\int_{\mathcal{Z}} \left(b_g \cdot \nabla f + \frac{1}{2} \text{tr} (D_g \nabla^2 f) + r_g f \right) \hat{\mu}_t^{g, N_g}(dz) \right] dt + dM_t^{g, f, N_g}, \quad (32)$$

where the martingale term is

$$dM_t^{g, f, N_g} = \frac{M_g^0}{N_g} \sum_{i=1}^{N_g} w_t^{g,i} \nabla f(Z_t^{g,i})^\top \sigma_g(Z_t^{g,i}, t; \hat{\mu}_t) dW_t^{g,i}. \quad (33)$$

Thus, the deterministic weak form in [equation \(27\)](#) and the formal reaction–drift–diffusion equation in [equation \(28\)](#) apply to the law-level measure μ_t^g , whereas the empirical particle approximation satisfies the same drift part plus a martingale fluctuation term.

7.10 Regularity conditions

For well-posedness of the law-level model and its particle approximation, we assume:

- (R1) the outputs of the three coefficient networks are measurable in (t, η) , where η denotes the finite-dimensional context argument, and are locally Lipschitz in z ;
- (R2) the context map depends on μ_t through a bounded finite-dimensional summary and is locally Lipschitz in that summary;
- (R3) $\sigma_{\min} > 0$ and $r_{\max} < \infty$ are fixed constants;
- (R4) whenever interpretable summaries ω and ϕ are used, they are bounded.

8 Explicit details behind the loss function

8.1 Why the loss must act on finite measures

The forward model predicts, for each perturbation g , one terminal finite measure

$$\mu_1^g.$$

The observed endpoint is also one finite measure

$$\nu_1^g = M_g^1 p_1^g.$$

Therefore the statistically coherent supervision compares

$$\mu_1^g \quad \text{to} \quad \nu_1^g$$

directly as finite measures.

If one instead splits supervision into:

- a normalized state-distribution loss on $\hat{p}_1^g$, and
- a separate scalar loss on $\hat{M}_g^1$,

one artificially decomposes one endpoint object into two losses. In the present formulation, terminal geometry and terminal abundance are fitted jointly by a single unbalanced discrepancy.

8.2 Derived endpoint summaries

Although the primary predicted object is μ_1^g , it is often useful to report

$$\hat{M}_g^1 = \mu_1^g(\mathcal{Z}), \quad \hat{p}_1^g = \frac{\mu_1^g}{\hat{M}_g^1}, \tag{34}$$

and, when ω is used,

$$\hat{q}_{g,1} = \frac{1}{\hat{M}_g^1} \int_{\mathcal{Z}} \omega(z) \mu_1^g(dz) \in \Delta^{K-1}. \tag{35}$$

These are summaries of the finite measure, not replacements for it.

8.3 Generalized KL divergence for finite measures

For finite nonnegative measures $\alpha, \beta \in \mathcal{M}_+(\mathcal{Z})$, define the generalized KL divergence by

$$\text{KL}(\alpha \parallel \beta) = \begin{cases} \int_{\mathcal{Z}} \log\left(\frac{d\alpha}{d\beta}\right) \alpha(dz) - \alpha(\mathcal{Z}) + \beta(\mathcal{Z}), & \alpha \ll \beta, \\ +\infty, & \text{otherwise.} \end{cases} \quad (36)$$

This is the correct KL notion when the compared objects need not have the same total mass.

8.4 Entropic unbalanced optimal transport

Let $c(z, z')$ be the latent ground cost, typically

$$c(z, z') = \|z - z'\|_2^2.$$

Because the OT loss is sensitive to latent geometry, this cost should be evaluated in a scaled or whitened latent space whenever the coordinates have strongly unequal variance. More general Mahalanobis or learned costs can also be substituted.

For finite measures $\alpha, \beta \in \mathcal{M}_+(\mathcal{Z})$, define

$$\begin{aligned} \text{UOT}_{\varepsilon, \tau}(\alpha, \beta) = \min_{\pi \in \mathcal{M}_+(\mathcal{Z} \times \mathcal{Z})} & \left\{ \int_{\mathcal{Z} \times \mathcal{Z}} c(z, z') \pi(dz, dz') + \varepsilon \text{KL}(\pi \parallel \alpha \otimes \beta) \right. \\ & \left. + \tau \text{KL}(\pi_1 \parallel \alpha) + \tau \text{KL}(\pi_2 \parallel \beta) \right\}, \end{aligned} \quad (37)$$

where π_1, π_2 are the marginals of π .

This objective has four parts:

- (i) the transport term penalizes moving mass too far in latent space;
- (ii) the entropic term stabilizes the problem numerically;
- (iii) the marginal KL terms allow creation or destruction of mass relative to strict balanced transport;
- (iv) the parameter τ controls how costly that mass variation is.

8.5 Debiased unbalanced Sinkhorn divergence

To reduce entropic bias, define

$$\text{SD}_{\varepsilon, \tau}^{\text{unb}}(\alpha, \beta) = \text{UOT}_{\varepsilon, \tau}(\alpha, \beta) - \frac{1}{2} \text{UOT}_{\varepsilon, \tau}(\alpha, \alpha) - \frac{1}{2} \text{UOT}_{\varepsilon, \tau}(\beta, \beta). \quad (38)$$

8.6 Primary endpoint loss

The model-level endpoint loss is

$$\mathcal{L}_{\text{end}} = \sum_{g \in \mathcal{G}} \alpha_g^{\text{end}} \text{SD}_{\varepsilon, \tau}^{\text{unb}}(\mu_1^g, \nu_1^g), \quad (39)$$

where α_g^{end} are user-chosen weights.

In computation, this loss is approximated by replacing μ_1^g with the particle measure $\hat{\mu}_1^{g,N_g}$:

$$\hat{\mathcal{L}}_{\text{end}} = \sum_{g \in \mathcal{G}} \alpha_g^{\text{end}} \text{SD}_{\varepsilon, \tau}^{\text{unb}} \left(\hat{\mu}_1^{g,N_g}, \nu_1^g \right). \quad (40)$$

This loss is the central reason to formulate the endpoint as a finite-measure problem. It jointly penalizes:

- mismatch in terminal state geometry,
- mismatch in total terminal mass,
- and mismatch in the tradeoff between latent movement and mass change.

8.7 Model-side growth versus loss-side mass relaxation

The reaction field r_g and the unbalanced OT relaxation parameter τ both influence how terminal mass mismatch is resolved, but they play different roles.

The field r_g belongs to the forward biological model: it determines how perturbation-specific mass evolves along the interval. By contrast, the KL marginal relaxation in [equation \(37\)](#) belongs to the endpoint discrepancy: it allows imperfect agreement between predicted and observed finite measures when guide-abundance calibration is noisy, when the finite-particle approximation is imperfect, or when strict balanced transport is too rigid.

Because only a single interval is observed, these two mechanisms are only partially identifiable from endpoints alone. This is one reason to bound and regularize r_g and to choose τ conservatively rather than allowing the loss to absorb all mass mismatch.

8.8 Optional auxiliary composition diagnostic

If one wants a weak interpretable diagnostic, one may define

$$\mathcal{L}_{\text{comp}}^{\text{aux}} = \sum_{g \in \mathcal{G}} \alpha_g^{\text{comp}} \text{KL} \left(q_{g,1}^{\text{obs}} \parallel \hat{q}_{g,1} \right). \quad (41)$$

Because $q_{g,1}^{\text{obs}}$ is derived from the same endpoint single-cell data that define p_1^g , this term is optional and not part of the primary supervision.

9 Regularization and identifiability

9.1 Embedding shrinkage

For non-control perturbations,

$$\mathcal{R}_{\text{emb}} = \sum_{g \in \mathcal{G} \setminus \mathcal{G}_{\text{ctrl}}} \|a_g\|^2. \quad (42)$$

If [equation \(11\)](#) is used, add

$$\sum_g \sum_{r \in \mathcal{R}_g} \|\delta_{gr}\|^2. \quad (43)$$

9.2 Perturbation-modulation regularization

Let

$$\tilde{p}_t^g = \frac{\mu_t^g}{\mu_t^g(\mathcal{Z})}$$

denote the normalized predicted population of perturbation g at time t . A natural penalty on perturbation-specific deviations from control is

$$\mathcal{R}_{\text{mod}} = \sum_{g \in \mathcal{G} \setminus \mathcal{G}_{\text{ctrl}}} \int_0^1 \mathbb{E}_{z \sim \tilde{p}_t^g} \left[\|B_b(u_t(z; \mu_t))a_g\|^2 + \|B_\sigma(u_t(z; \mu_t))a_g\|^2 + |B_r(u_t(z; \mu_t))a_g|^2 \right] dt. \quad (44)$$

9.3 Dispersion regularization

Because diffusion is only weakly identifiable from two endpoint snapshots, the model benefits from discouraging excessive stochastic spread:

$$\mathcal{R}_{\text{disp}} = \sum_{g \in \mathcal{G}} \int_0^1 \mathbb{E}_{z \sim \tilde{p}_t^g} \left[\|\sigma_g(z, t; \mu_t)\|_F^2 \right] dt. \quad (45)$$

9.4 Network parameter regularization

A generic parameter penalty is

$$\mathcal{R}_{\text{nn}} = \|\theta_b\|^2 + \|\theta_\sigma\|^2 + \|\theta_r\|^2 + \|\theta_c\|^2. \quad (46)$$

9.5 Why strong regularization is necessary

From only two state-resolved time points, drift, diffusion, growth, and context feedback are not equally identifiable. In practice:

- drift carries the main burden of explaining large-scale state change;
- diffusion should remain relatively simple and strongly regularized;
- growth must be bounded because it exponentiates into the particle weights;
- context effects should remain low-dimensional and well constrained.

These are not merely numerical choices; they are part of the effective interval modeling assumptions.

10 Full training objective

The primary objective is

$$\mathcal{L} = \lambda_{\text{end}} \mathcal{L}_{\text{end}} + \lambda_{\text{emb}} \mathcal{R}_{\text{emb}} + \lambda_{\text{mod}} \mathcal{R}_{\text{mod}} + \lambda_{\text{disp}} \mathcal{R}_{\text{disp}} + \lambda_{\text{nn}} \mathcal{R}_{\text{nn}} + \lambda_{\text{comp}}^{\text{aux}} \mathcal{L}_{\text{comp}}^{\text{aux}}. \quad (47)$$

The default main formulation takes

$$\lambda_{\text{comp}}^{\text{aux}} = 0$$

or very small.

In numerical training, $\mathcal{L}_{\text{end}}$ is replaced by its particle approximation $\hat{\mathcal{L}}_{\text{end}}$ from [equation \(40\)](#).

11 Numerical approximation and training

11.1 Euler–Maruyama approximation

Let

$$t_\ell = \ell \Delta t, \quad \Delta t = \frac{1}{L}, \quad \ell = 0, \dots, L-1.$$

Using [equations \(29\)](#) and [\(30\)](#), a standard Euler–Maruyama update is

$$Z_{\ell+1}^{g,i} = Z_\ell^{g,i} + b_g(Z_\ell^{g,i}, t_\ell; \hat{\mu}_{t_\ell}) \Delta t + \sigma_g(Z_\ell^{g,i}, t_\ell; \hat{\mu}_{t_\ell}) \sqrt{\Delta t} \xi_\ell^{g,i}, \quad (48)$$

$$w_{\ell+1}^{g,i} = w_\ell^{g,i} \exp\left(r_g(Z_\ell^{g,i}, t_\ell; \hat{\mu}_{t_\ell}) \Delta t\right), \quad (49)$$

where

$$\xi_\ell^{g,i} \sim \mathcal{N}(0, I_d).$$

11.2 Context update

If the interpretable context construction [equations \(14\)](#) to [\(16\)](#) is used, then at each step one recomputes

$$\hat{M}(t_\ell) = \sum_{g \in \mathcal{G}} \hat{\mu}_{t_\ell}^{g, N_g}(\mathcal{Z}) = \sum_{g \in \mathcal{G}} \frac{M_g^0}{N_g} \sum_{i=1}^{N_g} w_\ell^{g,i}, \quad (50)$$

$$\hat{q}_{t_\ell} = \frac{1}{\hat{M}(t_\ell)} \sum_{g \in \mathcal{G}} \frac{M_g^0}{N_g} \sum_{i=1}^{N_g} w_\ell^{g,i} \omega(Z_\ell^{g,i}), \quad (51)$$

$$\hat{s}_{t_\ell} = \frac{1}{\hat{M}(t_\ell)} \sum_{g \in \mathcal{G}} \frac{M_g^0}{N_g} \sum_{i=1}^{N_g} w_\ell^{g,i} \phi(Z_\ell^{g,i}), \quad (52)$$

and then sets

$$\hat{c}_{t_\ell} = \Psi_{\theta_c}(\hat{q}_{t_\ell}, \hat{s}_{t_\ell}). \quad (53)$$

If a different context map is used, one updates $\hat{c}_{t_\ell}$ according to that construction instead.

11.3 Discrete terminal measures

At the terminal step $t_L = 1$, the particle approximation to the predicted finite measure is

$$\hat{\mu}_1^{g, N_g} = \frac{M_g^0}{N_g} \sum_{i=1}^{N_g} w_L^{g,i} \delta_{Z_L^{g,i}}, \quad (54)$$

while the observed terminal measure is

$$\nu_1^g = \frac{M_g^1}{n_g^1} \sum_{j=1}^{n_g^1} \delta_{y_{gj}}. \quad (55)$$

The discrete unbalanced OT problem underlying [equation \(40\)](#) is then built from the cost matrix

$$C_{ij}^g = c(Z_L^{g,i}, y_{gj}).$$

11.4 Practical training loop

A practical training loop is:

- (1) construct the shared latent space $\mathcal{Z}$ from pooled $P4$ and $P60$ data;
- (2) choose normalized guide-abundance masses M_g^0, M_g^1 and, if needed, filter or pool perturbations with insufficient state support;
- (3) initialize particles at the observed $P4$ cells for each perturbation;
- (4) jointly simulate all perturbation-indexed populations from $t = 0$ to $t = 1$ using [equations \(48\)](#) and [\(49\)](#);
- (5) compute the terminal empirical measures $\hat{\mu}_1^{g, N_g}$ and the derived summaries $\hat{M}_g^1$, $\hat{p}_1^g$, and optionally $\hat{q}_{g,1}$;
- (6) evaluate the approximate endpoint loss [equation \(40\)](#) and the regularizers [equations \(42\)](#) and [\(44\)](#) to [\(46\)](#);
- (7) optionally evaluate [equation \(41\)](#) for diagnosis or weak auxiliary regularization;
- (8) update parameters by stochastic gradient descent.

If jointly simulating all perturbations is too costly, one may use a stratified minibatch of perturbations to estimate the mean-field terms. That changes the numerical estimator, not the population-level model.

12 Biological outputs and counterfactuals

The formulation produces the following outputs.

12.1 Predicted terminal finite measure

The central endpoint object is

$$\mu_1^g,$$

which jointly encodes terminal abundance and terminal state allocation. In practice, this object is accessed numerically through its particle approximation $\hat{\mu}_1^{g, N_g}$.

12.2 Predicted abundance effect

The abundance or selection effect of perturbation g is summarized by

$$\hat{M}_g^1 \quad \text{or} \quad \log \hat{M}_g^1 - \log M_g^0.$$

12.3 Predicted terminal state distribution

The normalized endpoint state distribution is

$$\hat{p}_1^g.$$

12.4 Predicted terminal coarse composition

If ω is used, the predicted coarse endpoint allocation is

$$\hat{q}_{g,1}.$$

This is an interpretive summary, not the main fitted target.

12.5 Intrinsic versus context-mediated effect

Because the model separates perturbation embedding from tissue context, one may compare:

- (a) a full simulation using the self-consistent mean field c_t , and
- (b) a simulation with c_t clamped to a reference trajectory or removed.

This gives a principled decomposition of endpoint effects into intrinsic perturbation components and context-mediated components.

12.6 Control-referenced perturbation effect

Because controls correspond to $a_g = 0$, one may also compare:

- (a) evolution under a_g , and
- (b) evolution under $a = 0$,

starting from the same initial state distribution and the same chosen context protocol. This directly quantifies the perturbation effect relative to the control continuation law.

13 What this model is, and is not

13.1 It is a mean-field interaction model

Mathematically, the model is a perturbation-indexed McKean–Vlasov reaction–drift–diffusion system with low-dimensional population feedback through the tissue-context vector c_t .

13.2 It is not a full mean-field game

A literal mean-field game would require explicit control variables, value functions, and a coupled Hamilton–Jacobi–Bellman / Fokker–Planck system. None of those are introduced here. The model uses mean-field feedback, but it is not a full equilibrium game.

13.3 It estimates an effective interval generator

From two state-resolved time points, the continuous-time law between them is not uniquely identifiable. The learned coefficient fields must therefore be interpreted as a regularized effective generator on $[0, 1]$, not as the unique microscopic biological truth.

13.4 It supports intervention-conditional interpretation, not unrestricted causal claims

The model is suitable for counterfactual simulation within the support of the observed perturbation-conditioned data and latent representation. It does not, by itself, identify unrestricted causal effects outside that support.

14 Concise summary

The formulation can be summarized compactly as follows:

For each perturbation g , start from the observed $P4$ finite measure $\mu_0^g = M_g^0 p_0^g$, define a law-level weighted McKean–Vlasov SDE in latent space whose drift, stochastic dispersion, and growth are modeled by three separate control-anchored neural networks conditioned on latent state, time, and low-dimensional tissue context, approximate that law numerically by a weighted particle system, and fit the resulting terminal finite measure μ_1^g to the observed $P60$ finite measure $\nu_1^g = M_g^1 p_1^g$ by a single unbalanced optimal-transport discrepancy.

Equivalently, the model has five layers:

- (1) a *biological mechanism layer*, which separates movement, stochastic spread, and abundance change;
- (2) a *law-level SDE layer*, which defines the reaction–drift–diffusion finite-measure dynamics;
- (3) a *particle approximation layer*, which provides a trainable Monte Carlo approximation to the law-level dynamics;
- (4) a *mean-field context layer*, which feeds back low-dimensional tissue information;
- (5) an *endpoint loss layer*, which compares predicted and observed terminal finite measures by unbalanced optimal transport.

This is the revised formal version of the $P4 \rightarrow P60$ control-anchored mean-field neural differential equation formulation after making the law-versus-particle distinction explicit, clarifying the role of mass proxies and unbalanced endpoint fitting, and preserving the biology-first presentation.